Midterm 2

Graded

Student

Aaron Xiong

Total Points

26 / 30 pts

Question 1

Question 1

10 / 10 pts

✔ **– 0 pts** Correct

– 2 pts Part (b) wrong

– 2 pts Part (c) incomplete/ error

– 2 pts Part (d) incorrect

– 1 pt Part (d) not enough

– 9 pts Little progress

– 10 pts No progress

Question 2

8 / 10 pts

– 0 pts Correct

– 10 pts Blank or Incorrect

– 2 pts Part A Incorrect

– 0.75 pts Part B) A correct thought about linear dependence but some issues in execution/producing the polynomial

– 1.5 pts Part B: Fails to bring up linear dependence of the correct vectors/generate a dependence relationship to reach a polynomial or correctly apply minimal polynomial, but has some ideas

– 2 pts Part B: Assumed some property such as v like being an eigenvector but still generates a polynomial.

– 3 pts Part B Incorrect (Common mistakes: assuming that p is given and trying to show $p(T) = 0$ instead of looking to find p; assuming p is given and trying to pick T to match it)

– 0.25 pts Minor Tweak: small operator equation issues, e.g. $p(T)v$ is an operator applied to a vector, $p(Tv)$ is not well defined

– 0.5 pts Part C) Minor Tweak: how do you know your eigenvector is nonzero? (when connecting $(T - \lambda I)q(T)v = 0$ to the eigenvector).

✔ **– 2 pts** Part C: Identified that a zero of the polynomial in B) is important, but failed to prove it is an eigenvalue/the connection to an eigenvector

+ 1 pt Part C: Partial credit on the eigenvector connection for factoring p but making operator equation errors

– 2 pts Part C

– 3 pts Part C: A major gap in reasoning

– 4 pts Part C) Some ideas, but little in the way of correct reasoning (Including some assumption about an eigenvalue or eigenvector existing. Notice, a minimal polynomial does not guarantee eigenvalues without the Fundamental Theorem of Algebra either)

– 5 pts Part C Incorrect

Question 3

8 / 10 pts

✔ – 0 pts Part (a): Correct

– 1 pt Part (a): Minor error(s)

– 2 pts Part (a): Correct idea but major error(s)

– 3 pts Part (a): Some relevant ideas

– 4 pts Part (a): Incorrect

– 0 pts Part (b): Correct

– 1 pt Part (b): Minor error(s)

✔ – 2 pts Part (b): Correct idea but major error(s)

– 3 pts Part (b): Some relevant ideas

– 4 pts Part (b): Incorrect

✔ – 0 pts Part (c): Correct

– 1 pt Part (c): Correct idea but major error(s)

– 2 pts Part (c): Incorrect

Math 110, Fall 2026 **Midterm Exam 2** **Prof. Hammond**

Friday, October 30, 2026

NAME: XIONG, AARON **SID:** 3041055980

LAST NAME, OTHER NAMES (IN BLOCK CAPITALS)

This is a 50 minute exam with a total of 30 points. There are three questions, worth 10 points each. Points per part are indicated in parentheses.

NOTATION

For a linear map T, its null space and range are denoted by $\operatorname{null} T$ and $\operatorname{range} T$. All bases are ordered when coordinates or matrices are requested.

INSTRUCTIONS

1. We will be scanning your exam and using Gradescope. Please write only within the black borders of each page, and write your name in the space provided on each question page.
2. Please write a concise answer to each part in the space provided, including calculations or reasoning as well as the final answer.
3. You may quote without proof any theorem or example from the text, lecture notes, or worksheets, provided the result is clearly stated, unless the problem asks you to prove it.
4. Extra space for each question is provided on the back of the preceding page. Use that page only for work on the indicated question.
5. The back of the final page may be used for rough work on any question. It will not be graded.
6. You are expected to do your own unaided work. Please read and sign the statement below.

I agree to abide by the Berkeley Honor Code. On my honor, I have neither given nor received any assistance while taking this exam.

Your signature: 

Extra space for QUESTION 1 ONLY

Name: _____

1. A subspace and its annihilator. Let $V = \mathbb{R}^3$ and

$$U = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

a) (2) What is meant by U^0, the annihilator of U in V'?

$U^0 \subseteq V' \quad U^0 = \{ \varphi : \varphi(u) = 0 \ \forall u \in U \}$

b) (3) Find a basis of U^0.

$\varphi \begin{pmatrix} a \\ a+b \\ b \end{pmatrix} = 0 \quad U^0 = \text{span}\{(1, -1, 1)\}$

$\varphi \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x - y + z$

c) (3) Extend the displayed ordered pair of vectors to a basis $\mathcal{B}$ of V and find its dual basis.

$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$

dual basis

$\varphi_1(v_i) = \begin{cases} 1 & \text{if } i=1 \\ 0 & \text{otherwise} \end{cases} \quad \varphi_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1 \Rightarrow \varphi_1 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{x - z + y}{2}$

$\varphi_2(v_i) = \begin{cases} 1 & \text{if } i=2 \\ 0 & \text{otherwise} \end{cases} \quad \varphi_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 1 \Rightarrow \varphi_2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{y - x + z}{2}$

$\varphi_3(v_i) = \begin{cases} 1 & \text{if } i=3 \\ 0 & \text{otherwise} \end{cases} \quad \varphi_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 1 \Rightarrow \varphi_3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{z - y + x}{2}$

d) (2) For the basis $\mathcal{B}$ that you chose in (c), explain why the third element of its dual basis spans U^0.

Because $\dim U^0 = \dim V - \dim U = 3 - 2 = 1$

any vector in U^0 is spanning

$\varphi_3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{x - y + z}{2} = \frac{1}{2}(1, -1, 1)$

$\Rightarrow \varphi_3 \in U^0$

Extra space for QUESTION 2 ONLY

Name: _____

2. Existence of complex eigenvalues. Let V be a nonzero finite-dimensional vector space over $\mathbb{C}$, and let $T : V \to V$ be linear.

a) (2) State the Fundamental Theorem of Algebra.

Every polynomial with real coefficients has a root in $\mathbb{R}$ and can be factored as $c(x-r_1)\cdots(x-r_n)$

b) (3) Let $v \in V$ be nonzero. Prove that there exists a nonconstant polynomial $p \in \mathbb{C}[z]$ such that $p(T)v = 0$.

Let $n = \dim V$.

We know the sequence $v, Tv, \cdots, T^n v$ is linearly dependent as there are $n+1 > n = \dim V$ vectors.

$\therefore \exists\, a_0, \cdots, a_n$ s.t $a_0 v + a_1 Tv + \cdots + a_n T^n v = 0$, $a_0, \cdots, a_n$ not all 0.

take $p(x) = a_0 + a_1 x + \cdots + a_n x^n$. Then $p(T)v = 0$.

c) (5) Using the Fundamental Theorem of Algebra and part (b), prove that T has an eigenvector.

$Tv = \lambda v$

$p(T)v = a_0 v + a_1 \lambda v + \cdots + a_n \lambda^n v = p(\lambda)v = 0$.

Then $p(\lambda)$ must be 0 since eigenvectors are nonzero by definition,

so this λ is a root of p, which we know exists by FTA.

By FTA, $\exists\, \lambda$ s.t $p(\lambda) = 0$.

$a_0 v + a_1 \lambda v + \cdots a_n \lambda^n v = p(\lambda) = 0 = p(T)v = a_0 v + a_1 Tv + \cdots + a_n T^n v$.

if $v \neq 0$

$\therefore \lambda v = Tv$ for $v \neq 0$, which is the definition for an eigenpair.

Extra space for QUESTION 3 ONLY

Name: _____

3. Eigenvalues and the minimal polynomial. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have matrix, in the standard basis,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

a) (4) Determine the eigenvalues of T and their corresponding eigenspaces.

$\lambda = 2$:

$$\left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right)$$

$y = 0$

$z = 0$

$E(2, T) = \text{span}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\lambda = -1$

$$\left(\begin{array}{ccc|c} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$3x + y = 0$

$3y = 0$

z is free

$E(-1, T) = \text{span}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

b) (4) Determine the minimal polynomial of T. Justify your answer.

$T(x, y, z) = (2x + y, 2y, -z)$

$T^2 = (4x + 4y, 4y, z)$ $\quad 4x + 8x, \; 4y, \; -2z$

$T^3 = (8x + 12y, 8y, -z)$ $\quad 4x + 2y, \; 4y, \; -2z$ ✓

$(x-2)^2(x+1) = (x^2 - 2x + 4)(x+1)$

$= (x^3 - 2x^2 + 4x + x^2 - 2x + 4) = x^3 - x^2 + 2x + 4$

c) (2) Is T diagonalizable? Explain.

No, T is not diagonalizable because

$$\text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\} \neq \mathbb{R}^3.$$

$\mathbb{R}^3$ is not a direct sum of the eigenspaces.

This blank page can be used for rough work on any question, but it will not be graded.